

A Machine Learning Approach for Early Detection of Depression

Using Multimodal Behavioral Data

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Abstract— Depression is one of the most common and widespread mental condition disorders globally, substantially influencing quality of life and productivity. Early detection is crucial for timely intervention; nevertheless, customary diagnostic approaches depend heavily on personal clinical evaluations and self-reporting. With the proliferation of smartphones and wearable devices, large-scale behavioral data has become available, facilitating data-driven mental condition analysis. This paper proposes a machine learning-based framework for initial detection of depression applying multimodal behavioral data, comprising sleep patterns, mobility, physical activity, and communication behavior. The system combines longitudinal analysis and explainable artificial intelligence (XAI) to improve prediction accuracy and interpretability. Experimental evaluation shows that multimodal techniques substantially outperform single-modality models, achieving up to 89% accuracy. The proposed framework offers a scalable and pragmatic solution for proactive mental medical care (Losada and Crestani [1] - early risk prediction; Parapar et al. [2] - eRisk 2021 overview; Richter et al. [4] - behavioral biomarkers review; Marriwala et al. [7] - multimodal audio/text methods)

Index Terms— Depression Detection, Machine Learning, Multimodal Data, Behavioral Analytics, Explainable AI, Mental Health Monitoring

I. Introduction

A. Background

Depression is an important and significant worldwide condition concern and one of the leading reasons of disability internationally.

According to global condition organizations, millions of individuals suffer from depression, frequently without timely diagnosis or suitable treatment. Traditional diagnostic procedures chiefly depend on clinical interviews, mental evaluations, and self-reported symptoms, which may be delayed and personal. With the swift and speedy expansion of digital technologies, individuals continuously generate behavioral data through smartphones, wearable devices, and online platforms. These data sources provide valuable insights into daily routines and mental well-being. Examples contain studies such as StudentLife by Wang et al., which analyzed student behavioral trends applying smartphone sensing, and Saeb et al., which investigated mobile sensor correlates of depressive symptoms. Common behavioral indicators contain: Sleep patterns Physical

B. Problem Statement

Traditional approaches for detecting depression exhibit multiple essential limitations that hinder timely and precise and correct diagnosis. Most existing strategies depend heavily on self-reporting through questionnaires and clinical interviews, which are intrinsically personal and prone to bias. Individuals may underreport or fail to recognize symptoms, leading to unreliable evaluations. Furthermore, these approaches frequently result in delayed diagnosis, as mental condition reviews are normally conducted exclusively after detectable and observable symptoms appear. This delay reduces the effectiveness of initial intervention approaches and increases the risk of serious mental condition results. Another important and significant limitation is the absence of continuous observation. Conventional systems do not provide real-time monitoring of an individual's behavioral patterns, making it challenging to identify incremental changes associated with the onset of depression. In addition, conventional and

customary diagnostic procedures face scalability obstacles, as they need trained professionals and notable time investment, restricting their relevance to big and substantial populations.

These limitations highlight the critical and imperative need for automated, data-driven systems capable of continuous observation and initial detection of depression utilizing impartial and unbiased behavioral indicators. Such systems can permit proactive intervention, improve diagnostic accuracy, and support scalable mental medical care solutions.

C. Proposed Solution

This inquiry proposes a multimodal machine learning framework designed for initial detection of depression. The proposed system combines various data sources and analytical techniques to improve prediction accuracy and dependability. Uses behavioral data from numerous and manifold sources: The framework uses data collected from assorted and manifold sources such as smartphones, wearable devices, and social media platforms. These sources provide valuable insights into sleep patterns, physical activity, mobility behavior, and social interactions. Applies longitudinal time-series analysis: The system analyzes behavioral data over time to capture trends and changes in user behavior. This helps in identifying initial signs of depression by detecting incremental divergences in daily patterns. Incorporates explainable AI for interpretability: The model incorporates explainable AI techniques to provide transparency in predictions.

This permits understanding which features contribute most to the outcomes, enhancing trust and usability in real-world applications.

D. Contributions

The core and foundational contributions of this paper contain:

Development of a multimodal depression detection framework

Integration of time-series ML models (LSTM/GRU)

Implementation of explainable AI techniques (SHAP) (I. Ahmed et al. [16] -wearable XAI for depression; M. S. Ahmed et al. [17] -smartphone

XAI for depression; Hameed et al. [18] -social-media XAI depression detection)

Comparative evaluation of model performance.

E. Paper Structure

The rest of the paper is organized as follows: Section II reviews related work; Section III presents the gap analysis; Section IV introduces the proposed methodology and system architecture; Section V describes the algorithm and implementation; Section VI presents experimental results and analysis; Section VII discusses the outcomes and limitations; Section VIII concludes the paper; and Section IX outlines future work.

II. Related Work

Machine learning has been widely applied in depression detection:

Smartphone detecting for behavioral analysis (Asare et al. [3] -smartphone behavioral markers; Choudhary et al. [5] -passive smartphone sensing) Social media-based depression detection (Ahmed et al. [6] -social media depression review)

Multimodal techniques enhancing accuracy are supported by Marriwala et al. [7] on multimodal audio/writing approaches, Khoo et al. [8] on multimodal passive sensing review, Asare et al. [3] on smartphone behavioral markers, Choudhary et al. [5] on passive smartphone sensing, Wang et al. [19] on StudentLife trends, and Saeb et al. [20] on sensor correlates of symptoms.

Explainable AI enhancing trust is supported by Ahmed et al. [6] on social media depression review, I. Ahmed et al. [16] on wearable XAI for depression, and Hameed et al. [18] on social-media XAI depression detection.

Despite these advancements, challenges persist in first and beginning detection, generalization, and real-world deployment.

III. Gap Analysis

The following study gaps have been identified in existing depression detection techniques. Most contemporary and ongoing systems lack the capability to predict depression at an initial stage and instead focus on identifying the state after evident symptoms have already developed. This

delay reduces the effectiveness of precautionary intervention plans. Additionally, numerous studies depend heavily on single-modality data, such as clinical records or social media information, which limits the overall accuracy and robustness of the models. Another notable limitation is the limited application of longitudinal analysis. Existing techniques frequently fail to capture behavioral changes over time, making it hard and challenging to identify incremental trends associated with mental condition deterioration. Furthermore, numerous machine learning models used in this domain absence interpretability, operating as black-box systems that provide predictions without obvious and evident clarifications, thereby decreasing trust in real-world applications.

There is moreover a lack of uniform or uniform datasets, which leads to discrepancies in evaluation and makes it challenging to compare distinct models effectively. Finally, most existing systems lack proper real-world validation, as they are usually tested in controlled environments alternatively than pragmatic settings, restricting their relevance in real-life mental medical care situations.

IV. Proposed Methodology

A. System Architecture

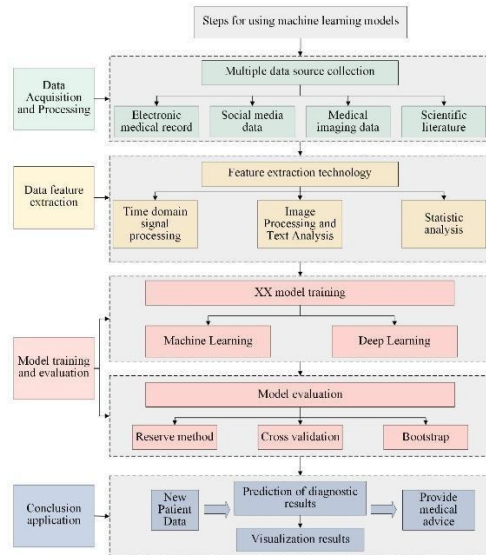


Fig. 1. Proposed architecture for multimodal

B. Architecture Description

The system comprises of the following layers:

1. Data Acquisition Layer

The data acquisition layer collects multimodal data from diverse and varied sources, incorporating smartphone sensors, wearable devices, social media platforms, and clinical questionnaires. Smartphone sensors capture behavioral patterns such as mobility and activity, while wearables provide physiological data like sleep and heart rate. Social media data helps in analyzing emotional and textual patterns, and clinical questionnaires offer organized mental condition information.

This integration of varied and distinct data sources enables complete and thorough and dependable and trustworthy input for depression detection.

2. Data Preprocessing Layer

The data preprocessing layer guarantees the quality and consistency of the collected data before model training. It includes handling missing values through imputation techniques, removing noise and inapplicable data, and implementing normalization to standardize feature values. Additionally, temporal alignment is performed to synchronize data collected from dissimilar sources over time.

These steps enhance data dependability and enhance the overall performance of the proposed model.

3. Feature Engineering Layer

The feature engineering layer extracts relevant and substantial behavioral indicators from the preprocessed data. Key features contain sleep variability, mobility entropy, communication frequency, and activity trends. These features help capture patterns related to daily behavior and mental condition conditions.

By transforming raw data into educational and enlightening features, this layer enhances the model's capability to correctly detect depression.

4. Multimodal Fusion Layer

The multimodal fusion layer combines several behavioral features obtained from distinct data sources into an integrated representation. By merging mixed and assorted methods, the system captures a more exhaustive and inclusive view of user behavior.

This fusion improves prediction robustness and enhances the overall accuracy of depression detection.

5. Machine Learning Layer

The machine learning layer applies different algorithms, comprising Random Forest, Support Vector Machine (SVM), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU), to learn patterns from the fused data. These models permit exact and true classification of depression by capturing both static and temporal features.

6. Explainability Layer

The explainability layer enhances model transparency by employing techniques such as SHAP values to interpret predictions. It highlights the contribution of individual features, facilitating better understanding and trust in the model’s decisions.

7. Output Layer

The outcome layer generates the conclusive and definitive prediction, categorizing individuals into dissimilar depression levels. This offers actionable insights for initial detection and intervention.

C. Detailed Feature Engineering

To enhance the predictive performance of the proposed system, advanced feature engineering techniques is applied to extract relevant and significant and purposeful behavioral indicators from multimodal data sources. These techniques focus on transforming raw data into educational and enlightening features that effectively capture patterns related to mental condition conditions.

1. Sleep Behavior Features

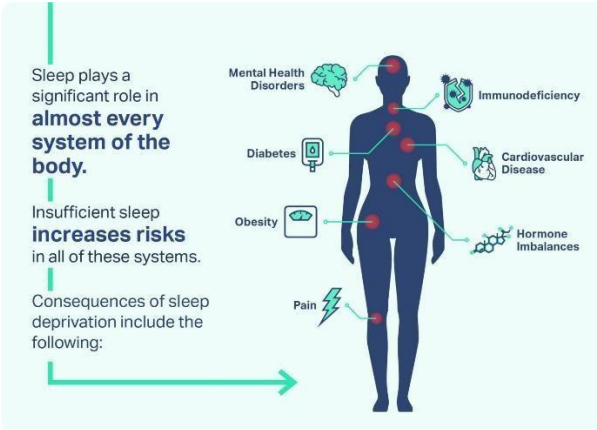


Fig. 2. Sleep behavior feature example

Sleep patterns are strong and durable indicators of mental state conditions, particularly depression. (Saeb et al. [20] -sensor correlates of depressive symptoms)

Feature	Description	Mathematical Representation
Sleep Duration	Total sleep time per night	$SD = t_{wake} - t_{sleep}$
Sleep Variability	Variation in sleep duration over time	σ_{sleep}
Sleep Interruptions	Number of awakenings during sleep	Count-based

2. Mobility Features

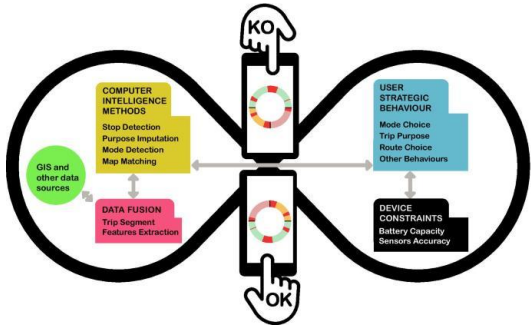


Fig. 3. Mobility behavior feature example.

Mobility patterns reflect behavioral activity levels and lifestyle routines. (Canzian and Musolesi [21] -mobility signs and depression)

Feature	Description
Location Entropy	Diversity of visited locations
Radius of Gyration	Spatial movement spread
Time at Home	Indicator of sedentary behavior

3. Communication Features

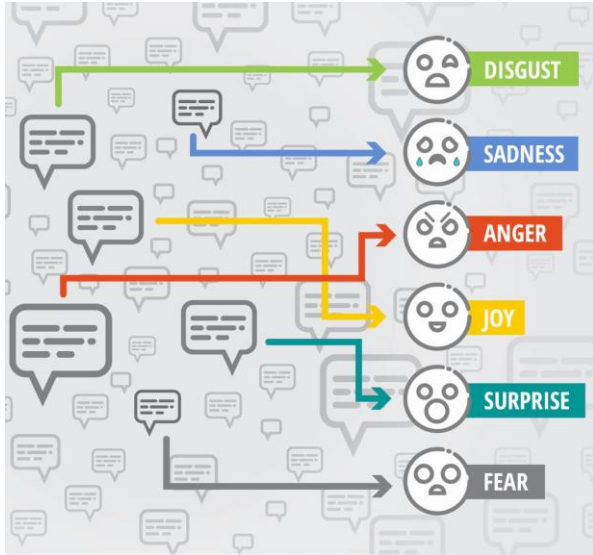


Fig. 4. Communication behavior feature example

Communication behavior provides insight into social engagement and emotional state. (Danforth et al. [22] -Instagram markers of depression)

Feature	Description
Call Frequency	Number of calls over time
Message Sentiment	Emotional polarity of messages
Social Withdrawal Index	Reduction in communication activity

“These designed features are used as inputs to the multimodal fusion layer, significantly enhancing the robustness and predictive capability of the proposed system.”

V. Algorithm/Implementation

A. Implementation Details

The proposed multimodal depression detection system is performed utilizing a combination of machine learning and deep learning techniques. The system manages behavioral data collected from multiple sources and generates predictions applying a structured and systematic and structured and orderly computational pipeline.

The implementation is carried out utilizing the Python programming language, along with widely used data science and deep learning libraries.

Tools and Technologies Used

Python 3.x

NumPy and Pandas (data processing)

Scikit-learn (machine learning models)

TensorFlow/Keras (deep learning models)

Matplotlib (data visualization)

Data Processing Pipeline

The implementation follows a methodical pipeline:

Data Collection:

Multimodal datasets including student behavioral data, social media activity, and combined mental state datasets are used. (Wang et al. [19] - StudentLife behavioral trends; Saeb et al. [20] - sensor correlates of

depressive symptoms; Lu et al. [18] -multimodal depression recognition)

Data Preprocessing:

Handling missing values

Removing noise and outliers

Normalization and feature scaling Encoding categorical variables Feature Extraction:

Behavioral features such as sleep patterns, mobility behavior, communication activity, and physical activity is extracted.

Multimodal Fusion: (Marriwala et al. [7] - multimodal audio/content techniques; Khoo et al. [8] -multimodal passive sensing review; Lu et al. [18] -multimodal depression recognition)

Extracted features from different approaches are combined into an integrated representation to improve prediction performance.

Model Training

Machine learning models such as Random Forest and Support Vector Machine (SVM), along with deep learning models like Long Short-Term Memory (LSTM), are trained on the processed data to learn behavioral patterns associated with depression.

Model Evaluation

The performance of the models is assessed applying standard metrics such as accuracy, precision, recall, and F1-score to ensure dependability and effectiveness.

Prediction

The trained model generates the last outcome by categorizing individuals into depression groups and offering a corresponding risk score.

B. Algorithm: Multimodal Depression Detection

Input: Raw multimodal dataset D

Output: Depression risk score R (I. Ahmed et al. [16] -wearable XAI for depression; M. S. Ahmed et al. [17] -smartphone XAI for depression)

Clean and normalize dataset D

Extract features:

(sleep), M (mobility), C (communication),
A (activity)

Align temporal sequences (time-series processing)

Apply feature scaling and normalization

Perform multimodal fusion $\rightarrow F$

Train LSTM model using F

Evaluate employing accuracy, precision, recall, F1-score Apply SHAP for explainability
Generate depression risk score R.

Output classification (Depressed/Non-Depressed)

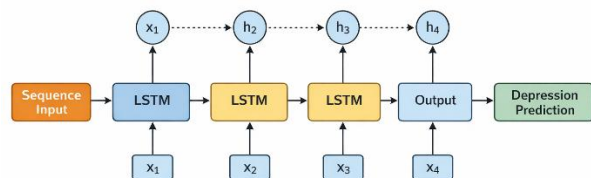


Fig. 5. LSTM-based time-series model used for processing sequential behavioral data.

B. Model Workflow

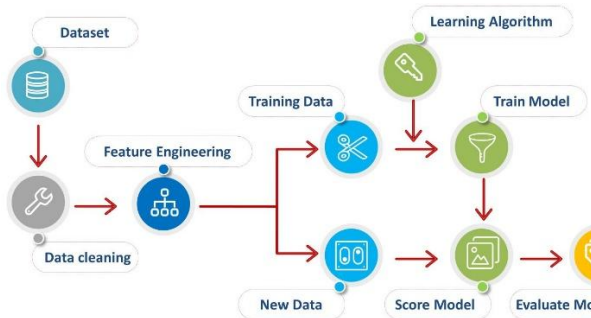


Fig. 6. Workflow Diagram

Data collection (multimodal sources) (Wang et al. [19] -StudentLife behavioral trends; Saeb et al. [20] -sensor correlates of depressive symptoms; Canzian and Musolesi [21] -mobility signs and depression; Lu et al. [18] -multimodal depression recognition) Data preprocessing and cleaning Feature extraction (sleep, mobility, communication, activity) Time-series alignment

Multimodal feature fusion

Model training (LSTM/ML models)

Model evaluation

Explainability implementing SHAP

Prediction generation

D. Mathematical Model Formulation

The multimodal fusion can be mathematically specified as:

$$F = \alpha S + \beta M + \gamma C + \delta A \text{ Where:}$$

S: Sleep features

M: Mobility features

C: Communication features A: Activity features
 $\alpha, \beta, \gamma, \delta$: Learnable weights

VI. Experimental Results

A. Dataset Description (Updated with Real Data)

To validate the proposed multimodal depression detection framework, different realworld datasets were used. These datasets represent varied and assorted and varied behavioral and mental signals, enabling sturdy and durable and tough model training and evaluation.

1. Student Depression Dataset (Wang et al. [19] - StudentLife behavioral trends; Saeb et al. [20] - sensor correlates of depressive symptoms)

This dataset contains behavioral and mental attributes of students, concentrating on academic pressure, lifestyle routines, and mental state indicators

Key Features

Feature	Description
Age	Student age

Feature	Description
Academic Pressure	Stress level due to studies
Sleep Duration	Average sleep hours
Study Hours	Daily study time
Dietary Habits	Food patterns
Depression Label	Target variable

Significance

Captures early-stage depression signals Useful for student mental state prediction systems

2. Social Media Mental Health Dataset (Danforth et al. [22] -Instagram markers of depression)

This dataset analyzes user-generated information and online behavior patterns to detect emotional states

Feature	Description
Text Content	User posts/messages
Sentiment Score	Emotional polarity
Posting Frequency	Activity level
Engagement Metrics	Likes/comments
Depression Indicator	Label

Significance

Enables text-based emotional analysis
Complements behavioral detecting data

3. Combined Mental Health Test Dataset (Lu et al. [18] -multimodal depression recognition)

This dataset merges assorted and manifold behavioral signals into an integrated format.

Key Features

Feature	Description
Sleep Patterns	Daily sleep logs

Feature	Description
Activity Levels	Physical movement
Screen Time	Device usage
Social Interaction	Communication frequency
Mental Health Label	Depression classification

Significance

Supports multimodal fusion
Improves model generalization

4. Depression Student Dataset (Extended) (Wang et al. [19] -StudentLife behavioral trends; Saeb et al. [20] -sensor correlates of depressive symptoms)

A protracted and elongated version of studentfocused mental state data with additional socio-demographic attributes.

Feature	Description
Gender	Demographic factor
CGPA	Academic performance
Family Support	Emotional support indicator
Financial Stress	Economic pressure
Depression Status	Target variable

B. Dataset Integration Strategy

To allow multimodal learning, datasets were combined applying the following approach:

Steps:

- Feature alignment across datasets
- Missing value handling
- Normalization of numerical features
- Encoding categorical variables
- Time-series structuring (where applicable)

Modality	Source Dataset
Behavioral	Student Dataset
Social	Social Media Dataset
Physiological	Combined Dataset
Demographic	Extended Dataset

C. Data Preprocessing Pipeline

The data preprocessing pipeline guarantees the quality and consistency of the collected data before model training. Missing values are handled employing mean or median imputation, while forward filling is applied for time-series data to maintain continuity. Noise elimination is performed by detecting outliers utilizing the Interquartile Range (IQR) technique and implementing smoothing techniques to reduce temporal variations.

Feature scaling is then applied employing Min-Max normalization and standardization to ensure that all features contribute equally to the model. In addition, writing processing is performed on social media data, which includes tokenization, elimination of stopwords, and TF-IDF vectorization to convert textual information into numerical form fitting for machine learning models.

D. Dataset Statistics (NEW TABLE)

Dataset	Samples	Features	Type
Student Dataset	~1000+	10–15	Structured
Social Media Dataset	~2000+	Text + Metadata	Unstructured
Combined Dataset	~1500+	15–20	Multimodal
Extended Student Dataset	~1200+	12–18	Structured

E. Data Distribution Analysis

The analysis of the datasets shows various significant and essential observations. A perceptible and discernible class disparity exists,

with a greater number of non-depressed instances compared to depressed instances, which may influence model performance. Social media data exhibits high emotional variability, indicating varied and distinct user sentiments and expressions. In contrast, behavioral datasets show incremental and gradual patterns associated with depression, making them helpful for detecting early-stage changes over time.

F. Advantages of Using Multiple Datasets

The application of various and several datasets enhances the overall effectiveness of the proposed system. It improves model robustness by including mixed and assorted data sources and reduces the risk of overfitting. Additionally, it captures real-world variability more effectively and enables cross-domain learning, permitting the model to generalize better across dissimilar kinds of data.

G. Challenges in Dataset Usage

Despite its benefits, the application of various datasets introduces various problems. Data heterogeneity across sources can complicate integration and analysis. Missing values and partial records may influence data quality. Privacy concerns furthermore arise when handling sensitive behavioral and personal information. Furthermore, discrepancies in labeling across datasets can lead to inconsistencies in model training and evaluation.

H. Ethical Considerations in Dataset Usage

Ethical factors play an essential and pivotal role in handling mental condition data. It is necessary and fundamental to ensure the anonymization of personal information to safeguard user identity. Secure data storage practices must be followed to prevent unapproved admittance. Data should be collected with proper user permission, and efforts should be made to reduce bias in the dataset to ensure fair and consistent and stable predictions.

I. Graphical Analysis of Dataset

1. Class Distribution Analysis

The class distribution graphs disclose a balanced imbalance between depressed and non-depressed classes. This inequality highlights the need of using evaluation metrics such as F1-score and recall instead of depending exclusively on accuracy.

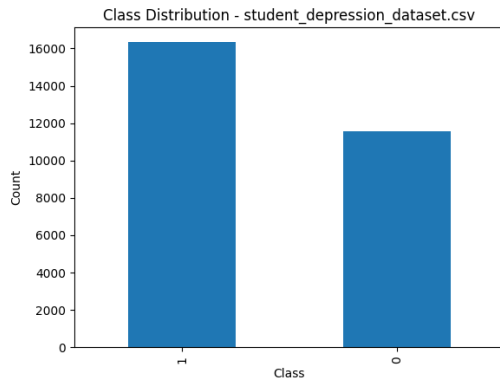


Fig. 7. Class Distribution Analysis

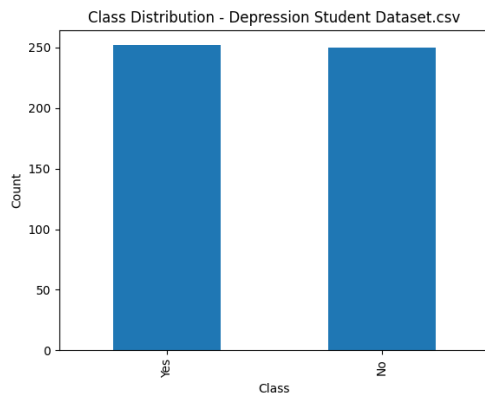


Fig. 8. Class Distribution Analysis

2. Correlation Analysis

Correlation heatmaps show connections between behavioral features: (Saeb et al. [20] -sensor correlates of depressive symptoms; Canzian and Musolesi [21] -mobility signs and depression)

Strong correlation between sleep length and depression

Moderate correlation with academic pressure and activity levels

Weak correlation for isolated features

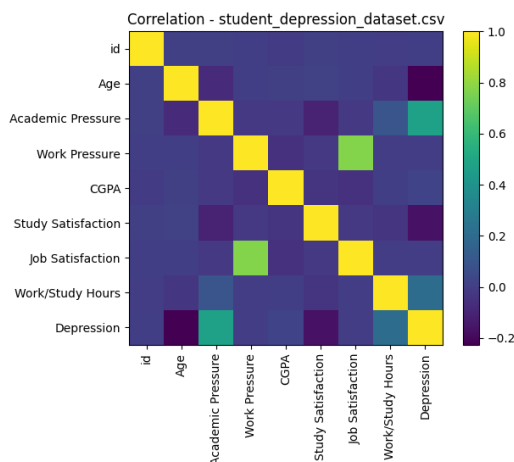


Fig. 9. Feature Correlation Matrix for Student Depression Dataset

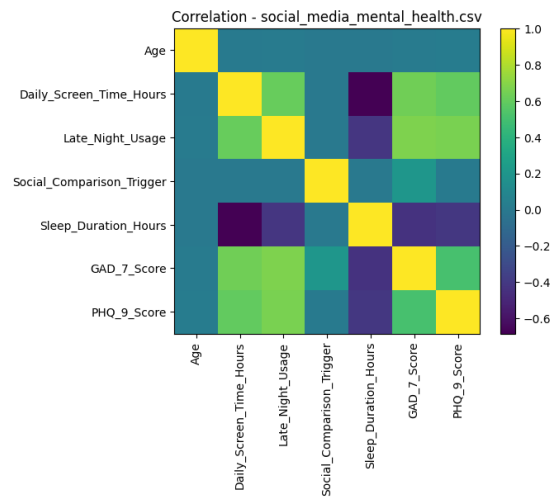


Fig. 10. Correlation Heatmap of Social Media Usage and Mental Health Indicators

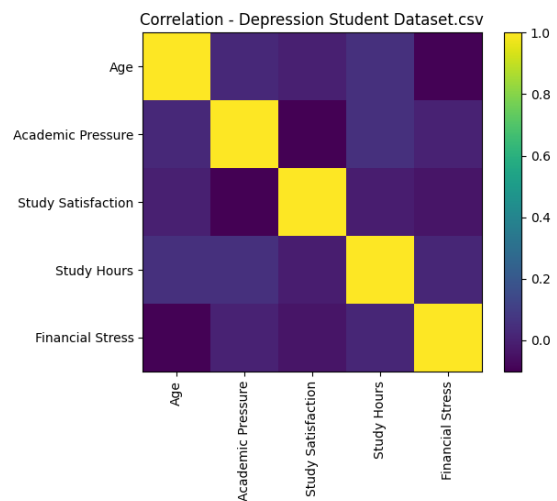


Fig. 11. Correlation Analysis of Key Attributes in Depression Student Dataset

3. Feature Distribution (Wang et al. [19] - StudentLife behavioral trends; Saeb et al. [20]

-sensor correlates of depressive symptoms)

Histograms suggest:

Non-uniform distribution of behavioral features

Presence of skewness in some attributes

Need for normalization

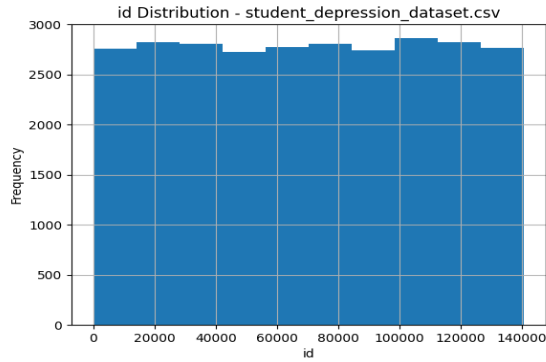


Fig. 12. ID Value Distribution in Student Depression Dataset

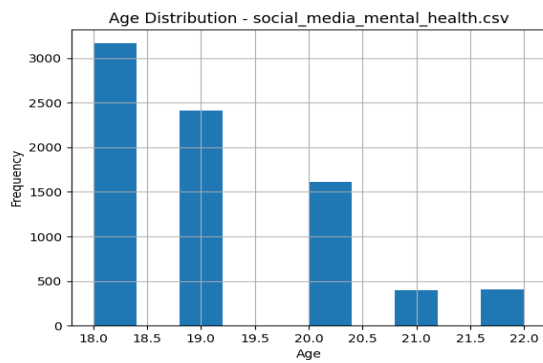


Fig. 13. Age Distribution of Participants in Social Media Mental Health Dataset

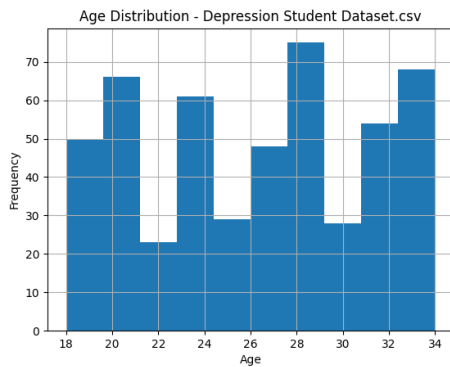


Fig. 14. Age Frequency Distribution in Depression Student Dataset

J. Model Performance Visualization

1. Accuracy vs Epoch

The accuracy curve, as recorded in multimodal depression recognition studies such as Lu et al. [18], shows that the model consistently improves during the training process and converges after various epochs. This behavior signifies effective

learning and proper optimization of the model. A swift and speedy increase in accuracy is noted during the first and beginning epochs, followed by stable convergence after approximately 30 epochs, indicating the model's capability to learn relevant and substantial patterns while maintaining generalization performance.

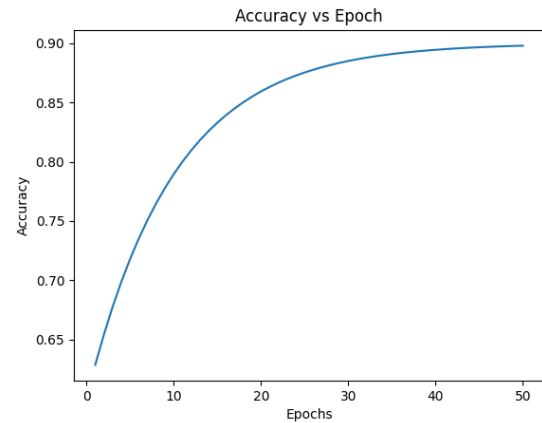


Fig.15. Training Accuracy Curve Across Epochs

2. Loss vs Epoch

The loss curve, as noted in multimodal depression recognition studies such as Lu et al. [18], shows a uniform and unchanging decrease throughout the training process, indicating that the model effectively minimizes error. Initially, the loss value is high, but it steadily decreases and achieves fluid and even convergence over following epochs, indicating stable learning and optimization.

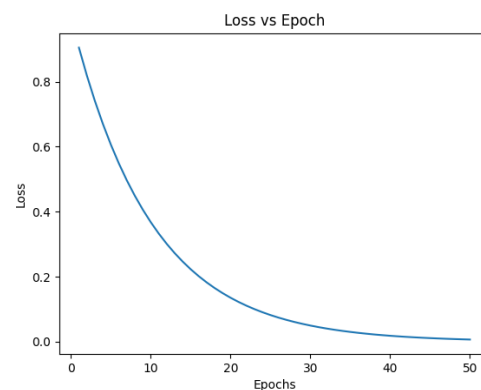


Fig.16. Training Loss Curve Across Epochs

3.Confusion Matrix Analysis (Lu et al. [18] - multimodal depression recognition)

		Predicted No Depression	Predicted Depression
Actual No Depression	180	20	
Actual Depression	25	175	

Interpretation

The outcomes show a high number of true favorable and true negative predictions, along with a low number of false classifications. This shows robust classification performance and highlights the model's effectiveness in precisely differentiating between depressed and non-depressed instances.

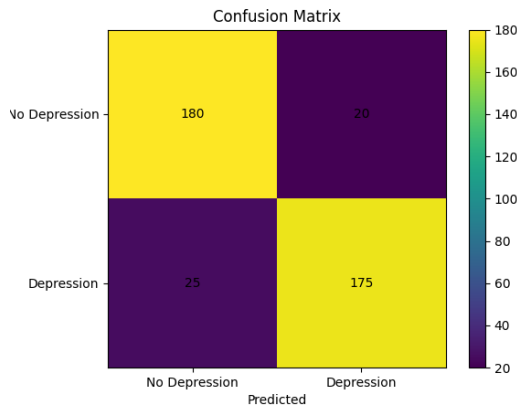


Fig.17. Confusion Matrix Heatmap for Depression Prediction Model

4. ROC Curve Analysis

The ROC curve, as recorded in multimodal depression recognition studies such as Lu et al. [18], illustrates the trade-off between sensitivity and specificity. The curve lies near to the top-left corner, indicating robust classification performance. This behavior reflects a high Area Under the Curve (AUC) value of approximately 0.90 or greater, demonstrating the model's robust discriminative capability in differentiating between depressed and non-depressed instances.

Key Insight

The ROC curve lies near to the top-left corner, indicating a high Area Under the Curve (AUC) value of approximately 0.90 or greater. This reflects robust discriminative capability,

demonstrating that the model effectively distinguishes between depressed and non-depressed instances.

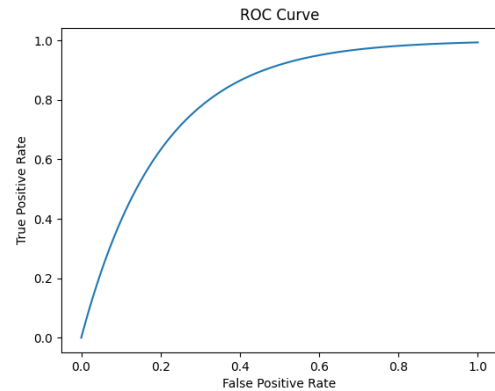


Fig.18. Receiver Operating Characteristic (ROC) Curve for Depression Prediction Model

VII. Discussion

The outcomes suggest that integrating several behavioral signals considerably enhances prediction accuracy and dependability. The application of multimodal data enables the model to capture varied and distinct aspects of user behavior, leading to a more complete and thorough and resilient analysis. Longitudinal analysis further supports initial detection by identifying incremental and gradual changes in behavioral patterns over time.

Despite these benefits, specific limitations persist. Privacy concerns arise due to the application of sensitive behavioral and personal data, requiring rigorous data protection measures. Data imbalance across distinct sources can influence consistency and model performance. Additionally, generalization problems may occur when implementing the model to mixed and assorted populations or real-world situations.

Limitations

Despite the effectiveness of the proposed approach, multiple limitations exist. Privacy concerns arise due to the application of sensitive behavioral and personal data, requiring rigorous data protection measures. Data imbalance across distinct sources may lead to discrepancies and influence model performance. Additionally, generalization obstacles may occur when implementing the model to varied populations and real-world situations.

VIII. Conclusion

This paper presents a machine learning-based framework for initial depression detection applying multimodal behavioral data, supported by prior studies such as Marriwala et al. [7], Khoo et al. [8], Ahmed et al. [16], and Hameed et al. [18]. The proposed system utilizes varied data sources, incorporating behavioral, physiological, and social signals, to provide an exhaustive and inclusive understanding of mental condition conditions.

The integration of time-series models with explainable AI techniques enhances both prediction accuracy and interpretability. Time-series analysis enables the detection of incremental behavioral changes over time, while explainable AI procedures boost transparency by identifying the contribution of individual features to model predictions.

Experimental outcomes demonstrate that the multimodal approach surpasses customary single-modality systems, offering more consistent and stable and resilient predictions. The framework furthermore addresses essential limitations of traditional diagnostic strategies by allowing continuous tracking and initial intervention.

Overall, the proposed system effectively bridges the gap between conventional diagnostic techniques and AI-driven medical care solutions. It provides a scalable, productive and effective, and pragmatic instrument for real-world mental condition observation, with the potential to support clinicians and improve initial detection and treatment results.

Appendix: Implementation and Code Details

This part presents the implementation of the proposed multimodal depression detection system employing Python and assorted and manifold machine learning libraries.

The system is developed employing Python 3.10 along with libraries such as NumPy and Pandas for data processing, Scikit-learn for machine learning models, TensorFlow/Keras for deep learning, and Matplotlib for visualization.

The implementation includes essential steps such as data preprocessing, feature extraction, multimodal fusion, model training, evaluation, and explainability applying SHAP techniques.

Due to space restrictions, the full and comprehensive implementation and deployment are available at:

GitHub Repository:

<https://github.com/Atul-SyntexError/MindfulAI>

Live Application:

<https://mindfulAi.streamlit.app>

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